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Inverse Problems and Imaging (02624)

Week 4

Tikhonov regularization: Optimization

Compact operator $K \in B(X, Y)$; X, Y are HS.

Equation with perturbation (noise) in data

$$Kx \approx y^\delta = y + d, \quad \|d\| < \delta.$$

Do not solve least squares problem

$$x_0 = \operatorname{argmin}_{x \in X} \|Kx - y^\delta\|^2.$$

Instead, for $\alpha > 0$ consider

$$x_\alpha = \operatorname{argmin}_{x \in X} \|Kx - y^\delta\|^2 + \alpha \|x\|^2.$$

Well-posed problem for any $y_\delta \in Y$, a unique solution

$$x_\alpha = R_\alpha y^\delta$$

Tikhonov regularization: Normal equation

The least squares optimization problem

$$\operatorname{argmin}_{x \in X} \|Kx - y^\delta\|^2$$

leads to normal equation and pseudo-inverse

$$(K^*K)x = K^*y^\delta \quad \Leftrightarrow \quad x = (K^*K)^{-1}K^*y^\delta = K^\dagger y^\delta.$$

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The Tikhonov problem

$$\operatorname{argmin}_{x \in X} (\|Kx - y^\delta\|^2 + \alpha\|x\|^2)$$

leads to the normal equation and regularized inverse

$$(K^*K + \alpha)x_\alpha = K^*y^\delta \quad \Leftrightarrow \quad x_\alpha = (K^*K + \alpha)^{-1}K^*y^\delta = R_\alpha y^\delta.$$

Tikhonov regularization: Filter factors

Exact reconstruction (bad idea)

$$x = \sum_j \frac{(y^\delta, y_j)}{\mu_j} x_j$$

Regularization via filtered versions

$$x_\alpha = \sum_j q_\alpha(\mu_j) \frac{(y^\delta, y_j)}{\mu_j} x_j$$

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$$\text{TSVD} \quad x_\alpha = \sum_{\mu_j^2 > \alpha} \frac{(y^\delta, y_j)}{\mu_j} x_j, \quad q_\alpha(\mu) = \begin{cases} 1 & \mu^2 > \alpha \\ 0 & \mu^2 \leq \alpha \end{cases}$$

$$\text{Tikhonov} \quad x_\alpha = \sum_j \frac{\mu_j^2}{\mu_j^2 + \alpha} \frac{(y^\delta, y_j)}{\mu_j} x_j, \quad q_\alpha(\mu) = \frac{\mu^2}{\mu^2 + \alpha}.$$

Matlab implementation

Rewrite for $K = A \in \mathbb{R}^{n \times m}$ and $y \in \mathbb{R}^n$

$$\|Ax - y\|_2^2 + \alpha \|x\|_2^2 = \left\| \begin{bmatrix} A \\ \sqrt{\alpha} I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2.$$

Implement in Matlab by

```
Aalpha = [A; sqrt(alpha)*eye(n)];  
bext = [bnoise; zeros(N,1)];  
xalpha = Aalpha \ bext;
```